SHRIANSH MANHAS

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I bring an excellent blend of expertise in **LLM** model **training**, retrieval-augmented generation (**RAG**), and data analytics. My experience spans **optimizing AI algorithms**, **refining ETL pipelines** to support scalable analytics workflows, and securing sensitive data through advanced encryption techniques. I have worked on **designing and implementing RAG-based system**s to enhance contextual relevance and response accuracy in LLMs. I have **co-authored research papers advancing data security strategies in AI-driven environments**. With a proven ability to communicate complex ideas clearly and collaborate across disciplines, I am well-equipped to **contribute to teams focused on cutting-edge LLM development, data-centric innovation** and cloud applications.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY — Atlanta, Georgia
Masters of Science (MS) Computer Science — **GPA:** 3.8/4.0 Jan 2026

NATIONAL INSTITUTE OF TECHNOLOGY, DELHI — Delhi, India
Bachelor of Technology (B. Tech) Computer Science and Engineering — **GPA**: 8.12 May 2024

Co-authored two research publications

INTERESTS AND SKILLS

Interests: Large Language Models (LLMs), Small Language Models (SLMs), Retrieval-Augmented Generation (RAG), Machine Learning, Data Mining, Model Security, Distributed Systems, Cloud Architecture, Scalable Data Pipelines
ML Ops & Data Engineering: Model Serving & Deployment (FastAPI), Pipeline Orchestration (Airflow / N8N), Distributed Training, Cloud Data Lakes, Feature Stores, Data Streaming (Kafka), locally quantized GPT-2
Special Focus: LLM Alignment & Safety, Prompt Injection Defense, Model Compression & Quantization, Efficient Fine-Tuning (LoRA/Adapters), Secure Multi-Agent Workflows, Graph Neural Networks(GraphSAGE), Small Language Models
Visualization & Analysis: Tableau, Power BI, Matplotlib, Seaborn
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, PEFT (**LoRA/QLoRA**), LangChain, **LangGraph**, DeepSpeed, **vLLM**, Pandas, NumPy, scikit-learn, DGL
Languages: Python, C++, CUDA, Java, SQL, Bash
Tools & Platforms: AWS, Azure, Docker, Kubernetes, Jenkins, Spark, Ray, Linux, Triton, N8N

WORK EXPERIENCE

AT&T — Alpharetta, USA
Context Engineer — March 2026-Present
- ***Supervisor***: Arthur Dileo
- Engineered 'OptiPricer,' an enterprise agentic data pipeline integrated within AT&T's core web interface and Salesforce, utilizing Kubernetes (K8s) clusters for seamless orchestration and high-availability scaling.
- Architected a dynamic multi-agent pipeline using FastAPI and Pydantic to ingest, parse, and structure heterogeneous unstructured and structured mobility data into strictly validated JSON schemas.
- Accelerated data onboarding velocity by 30% and eliminated parsing errors for mid-market mobility sellers by decoupling schema enforcement from raw ingestion, ensuring 84.3% downstream data compliance.

SP TECH (Creating growth-driven digital environments powered by Salesforce) — Atlanta, USA
Junior AI Developer — May 2025 – Aug 2025
- ***Supervisor***: Mr. Neeraj Parikh
- **Led a team of 3 interns**, completing 100% of sprint deliverables on time, improving release velocity by 30%.
- Developed an **MCP server for a context-aware Slack bot** for summarization, availability checks, smart scheduling, and real-time Q&A—integrated enterprise-grade security (OAuth 2.0, RBAC, channel isolation) with a scalable PostgreSQL infrastructure hosted on Glama. Achieved **p95 latency of 320 ms**, **<$0.002 cost/request**, and **≈92% retrieval accuracy** via **OpenAI-embedding–based semantic search**. Deployed using **FastAPI, Docker, and GitHub Actions CI/CD** for reproducible builds and cloud scalability. Cut stale query responses by 40% by implementing **a self-refreshing RAG pipeline**.

SKIT.AI (Conversational voice AI solution provider in the accounts and receivables Industry) — Bangalore, India
Software Developer Intern — May 2023 – Aug 2023
- ***Supervisor***: Mr. Akshay Deshraj
- Built **ETL functionality** in the Docker pipeline to insert custom datasets for testing the LLM architecture instead of doing the train-test split, thus making debugging easier for MLOps in an Agile environment
- Adding a fork from the component reduced error detection time by 16% for the NLP system

- Achieved a 12% increase in processing efficiency, enabling 20% more concurrent experiments and unlocking $8,000+/quarter in infrastructure savings.
- Presented a Tableau logical data model for pipeline enhancements to the CTO, influencing roadmap discussions and adoption of improved data practices.

INDIAN INSTITUTE OF TECHNOLOGY (IIT) ROORKEE Roorkee, India
Research Intern Jun 2022 – Aug 2022

- ***Topic***: Medical image security and authenticity via dual encryption; ***Supervisor***: Prof R. Balasubramanian, CSE Dept
- Proposed a novel dual encryption applying Blowfish and Signcryption algorithm in a certificateless generalized form
- The work resulted in a co-authored paper on Medical Image Security and Authenticity via Dual Encryption

UNIVERSITY PROJECTS

TERNO-AI(LINK)

- Database intelligence layer designed for Security and Accuracy, bridging the gap between AI Agents and Enterprise Data.
- Provides a unified, secure API for interacting with warehouse-scale databases while enforcing strict access controls and optimizing the database schema context for LLMs.

CYCLIC PRECISION OPTIMIZATION(LINK)

- Designed and developed a fully autonomous AI Dungeon Master for Dungeons & Dragons campaigns.
- GPT-2 Fine-Tuning on SQuAD. How do I increase Fine Tuning efficiency of a model while ensuring it remains light weight
- Trained the Model using LoRA, frozen weights. Evaluation compares Cyclic Precision to multiple forms of Dynamic Quantization.
- Resulted in a 14.4% increase in F1, 15% increase to EM with the tradeoff being Efficiency drop of 4.7%.

DND-DUNGEON MASTER (link)

- Designed and developed a fully autonomous AI Dungeon Master for Dungeons & Dragons campaigns.
- Focused on procedural world generation, dynamic story arcs, and real-time interaction.
- Leveraged a **locally quantized Qwen-7B** model deployed through a custom RAG pipeline.
- Integrated vector database (PostgreSQL) for efficient semantic search of in-game lore and prior events.

CACHE COHERENCE

- A multi-level Cache Simulation project with options to enable Victim/L2 settings to analyze system performance trade-offs.
- The simulator models real-world memory hierarchies and supports detailed tracking of hit/miss rates, block replacement policies, and coherence state transitions across levels.

OPTIMAL DECISION TREE FOR PACKET CLASSIFICATION (link)

- Implemented an open-source Genetic Algorithm to evolve decision tree structures. Represent trees as genomes, and implement genetic operators like crossover and mutation to generate efficient tree structures over iterations
- Outperforms prefix-based decision trees matches against packet header fields

mRNA DEGRADATION(COVID-19) (LINK)

- Built a hybrid deep learning architecture integrating **Graph Convolutional Networks (GCN), GraphSAGE, and GRUs** to predict RNA degradation patterns.
- Leveraged graph embeddings to capture nucleotide connectivity and sequence dynamics for temporal dependencies.
- Implemented k-fold cross-validation and data augmentation for robust generalization on the RNA structural dataset.
- Optimized training with AdamW and custom learning rate schedulers, improving predictive performance on the leaderboard.

DETECTING AI-GENERATED SCIENTIFIC PAPERS(IBM PROJECT) (link)

- Employed Bidirectional Transformers model (BERT) for NLP
- Modified the model by pruning the unnecessary layers using magnitude pruning, zeroing out the non-significant weights, and fitting the dataset, which was tokenized, cleaned, and preprocessed
- This model broke into the top 5 of a Kaggle competition

ADVERSARIAL PERTURBATIONS (link)

- Detect adversarial attacks in applications such as self-driving cars and most computer vision-based AI models
- Developed a model to detect attacks using FGSM on CIFAR10, PGD on SVHN, and Deep learning
- Achieving a high accuracy rate of 82.3% showcasing my problem-solving skills

PUBLICATIONS

- Kishore Babu Nampalle, **Shriansh Manhas**, Balasubramanian Raman, “Medical image security and authenticity via dual encryption,” Appl Intell. (Springer) 53, 20647–20659 (2023). (link)
- Ankur Kumar Singhal, **Shriansh Manhas**, Anurag Singh, “Beyond the traditional models: a network reconstruction based model for predicting and analysing individual health status,” Computing 107 (Springer), 14 (Nov 2024) (link)

CERTIFICATES (Coursera)

1. The foundational course on **Machine Learning by Andrew Ng, Stanford University** (link) **2. Probability Theory,** Statistics and Exploratory Data Analysis **by Ilya V. Schurov, HSE** (link) **3. Deep Learning Specialization** (link)